

Advanced Microeconomics

Dr. Juliane Hennecke
Nils Sulzmann

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Problem Set 4: Mathematical solutions

1) The utility function of an individual is $U(x_1, x_2) = \ln x_1 + \ln x_2$. The prices in period 0 are $p_1^0 = 1$ and $p_2^0 = 4$, the income is $w = 200$.

- a) Calculate the Marshallian demand functions, the indirect utility function, the expenditure function and the Hicks'ian demand functions.

Solution:

$$x_1(p_1, p_2, w) = \frac{w}{2p_1}$$

$$x_2(p_1, p_2, w) = \frac{w}{2p_2}$$

$$v(p_1, p_2, w) = \ln\left(\frac{w}{2p_1}\right) + \ln\left(\frac{w}{2p_2}\right) = \ln\left(\frac{w^2}{4p_1p_2}\right)$$

$$e(p_1, p_2, w) = 2\sqrt{p_1p_2}e^u$$

$$h_1(p_1, p_2, u) = \sqrt{\frac{p_2}{p_1}}e^u$$

$$h_2(p_1, p_2, u) = \sqrt{\frac{p_1}{p_2}}e^u$$

- b) Suppose the government introduces a tax t on good x_1 so that $p_1^1 = 2$. Calculate the Compensating Variation (CV) and the Equivalent Variation (EV) and interpret your results.

Solution:

$$EV = -58.579$$

$$CV = -82.843$$

- c) Calculate the “excess burden of taxation” associated with the new tax on good x_1 as $|EV| - t \times h(p^1, U^1)$, where $|EV|$ is the absolute value of the Equivalent Variation from (b), t is the tax, $h()$ the Hicksian demand, p^1 the new price level and U^1 the utility with the new price level. Why does this excess burden of taxation arise?

Solution:

$$EB = 8.579$$

2) The utility function of an individual is given by $u(x_1, x_2) = x_1^{\frac{1}{4}} x_2^{\frac{3}{4}}$. The wealth of the individual and the prices of good 1 and 2 are greater than 0 ($w, p_1, p_2 > 0$).

a) Determine the Marshallian demand functions.

Solution:

$$x_1 = \frac{w}{4p_1}$$

$$x_2 = \frac{3w}{4p_2}$$

b) Determine the indirect utility function.

Solution:

$$v_{p_1, p_2, w} = \left(\frac{w}{4p_1} \right)^{\frac{1}{4}} \left(\frac{3w}{4p_2} \right)^{\frac{3}{4}}$$

c) Determine the expenditure function and the Hicksian demand functions.

Solution:

$$e_{p_1, p_2, u} = \frac{u}{\left(\frac{1}{4p_1} \right)^{\frac{1}{4}} \left(\frac{3}{4p_2} \right)^{\frac{3}{4}}}$$

$$h_1 = u \left(\frac{p_2}{3p_1} \right)^{\frac{3}{4}}$$

$$h_2 = u \left(\frac{3p_1}{p_2} \right)^{\frac{1}{4}}$$